Setup for Material Jetting 350

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Class Outline

Material Jetting
Material Jetting Methods
Machine Components
Software for Material Jetting
Review: Methods, Components, and Software
Build Orientation
Material Selection
Material Delivery System Calibration
Build Platform Setup and Calibration
Review: Setup and Calibration
Post Processing Setup
Material Jetting Safety
Final Review

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Material Jetting

Material jetting, an additive manufacturing (AM) process, deposits layers of photoreactive resin, wax, or other materials from a print head. Once deposited, photoreactive resin build materials cure and harden under ultraviolet light (UV light) to form three-dimensional (3D) parts. Other build materials, such as nanoparticle materials, are cured using heat, while wax materials simply cool and harden, usually to form support structures.

Most material jetting machines have a three-axis configuration, though some advanced machines may rotate in a fourth axis. Generally, the print head moves on a gantry along the horizontal X and Y axes, and the build platform moves along the vertical Z axis. Some machines have a print head spanning the width of the build area that only moves horizontally along the X axis.

Setup for material jetting machines involves a range of tasks, including exchanging resin tanks or material cartridges, calibrating equipment, running test prints, and following all machine safety guidelines.

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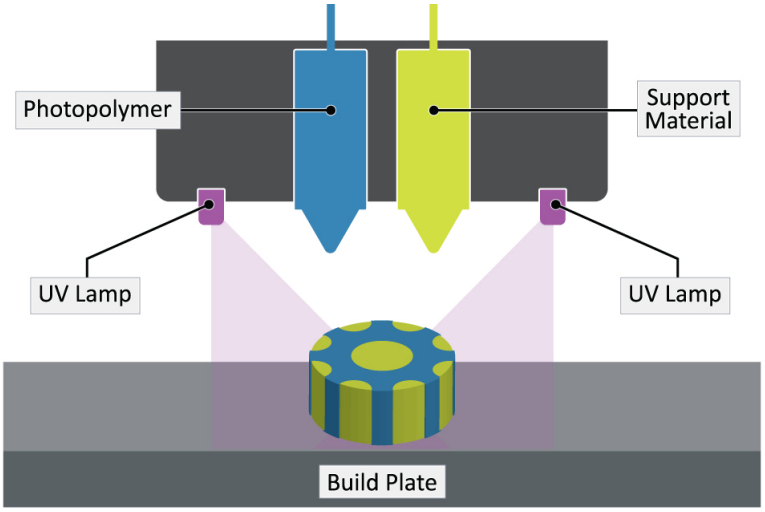
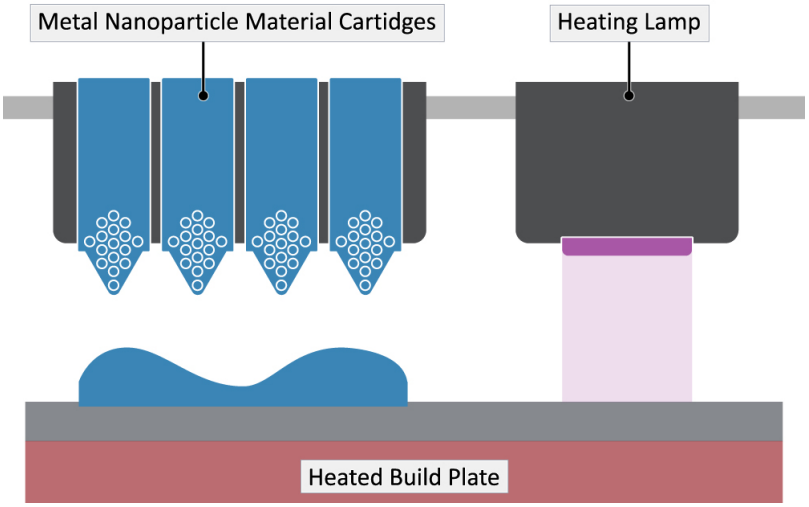
Material Jetting Methods



Photopolymer material jetting, the most common material jetting process, uses multiple print heads that dispense layers of photopolymer resins that cure and harden under a UV lamp. These machines may also use wax for support structures or as a separate build material. This method is ideal for prototyping but may also be used to build some end-use products, such as dental fixtures. Setup may involve managing material tanks, loading multiple materials, and adjusting colors in the software for multicolored builds.



Nanoparticle jetting (NPJ), an advanced process, uses print heads like standard machines but deposits ceramic or metal and support materials suspended in a carrier liquid. A solid part forms when the liquid evaporates. NPJ setup tasks are similar to photopolymer jetting but also involve adjusting a heating lamp and a heated build plate for the materials to dispense into layers accurately and to properly cure.

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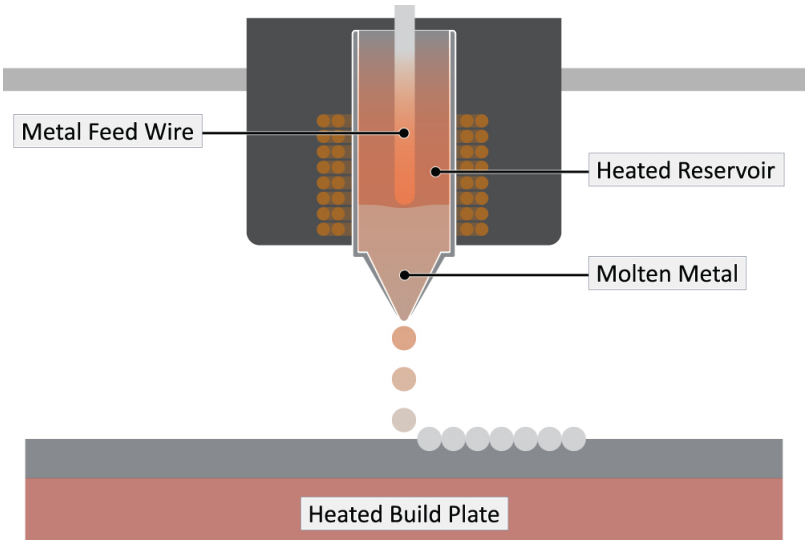

Another advanced method, liquid metal jetting (LMJ), deposits molten metal onto a heated build plate. The molten material is created from a metal wire that is melted in a heated reservoir. Heating elements and the wire feed system must be properly maintained and calibrated. Additionally, metal part production requires different setup considerations than the resins used in traditional material jetting, such as preparing for different post-processing steps. For instance, metal AM parts often require heat treating and cooling processes such as quenching.

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Machine Components

There are many components to consider when setting up material jetting machines.

The material delivery system should include properly installed material cartridges free from leaks, which ensures the material is correctly transported to the print head.

The print head contains hundreds to thousands of nozzles that deposit droplets of material in quantities precisely controlled by the print software. Each nozzle deposits material using pneumatic pressure or piezoelectric actuation. Print head nozzles should be cleaned and calibrated before use.

The UV lamp for curing photopolymer resins may require cleaning or adjustment during setup. For other material types, you may need to check and adjust heating elements for proper curing.

The leveling roller passes over each part layer to ensure the material is level. Leveling rollers may need to be installed, cleaned, or adjusted during setup.

The build platform supports the printed part, while the elevator raises and lowers it along the Z axis. Setup may include cleaning or leveling the platform and verifying the elevator works properly.

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Software for Material Jetting

Like all AM methods, material jetting machines use converted data from computer-aided design (CAD) software. In the CAD software, engineers determine the geometry, components, materials, colors, and overall function of a part.

Specialized printing programs convert the design files into layers and printing instructions for use in a material jetting machine. These instructions include automated machine functions and mapped toolpaths for building the part.

Before creating the build file, you may need to make manual adjustments in the printing program, such as changing colors or materials, adjusting print resolution, or changing the print speed.

Once the operator finalizes any selections, setup is complete, and the operator runs the print program to build the part. However, you should monitor the printing program, especially for new builds, to ensure the printing process works as intended. Faulty prints require you to return to the setup process for troubleshooting.

Figure 1:

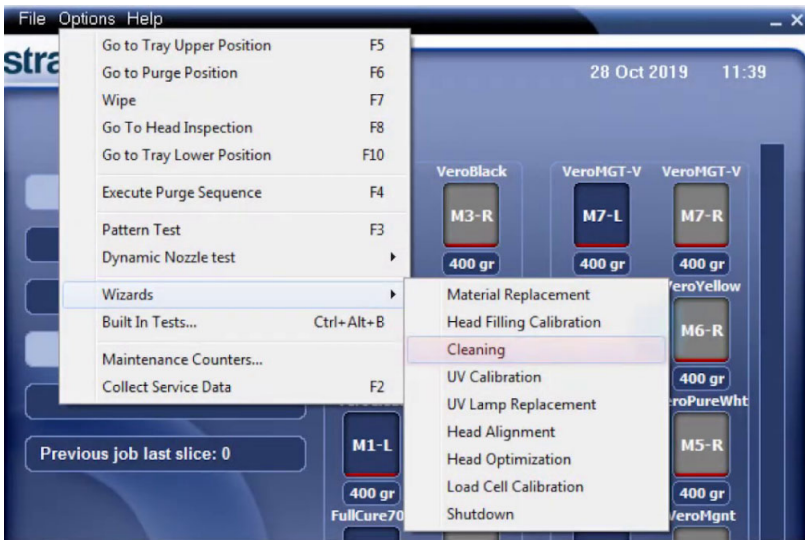

Figure 2: Photo courtesy of Stratasys

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Review: Methods, Components, and Software

1. Most material jetting machines use a five- or six-axis configuration. **FALSE**
2. Nanoparticle jetting is the most common material jetting method. **FALSE**
3. The leveling roller may need to be installed and adjusted during material jetting setup. **TRUE**
4. Operators can adjust print resolution settings in the printing software. **TRUE**

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Build Orientation

Another key software setup task for material jetting may involve adjusting the build orientation. Like other three-axis methods, material jetting requires support materials to create overhangs and other complex features. Proper build orientation minimizes the amount of build materials required while also optimizing the total printing time.

Additionally, material jetting machines can build multiple copies of the same part or multiple components at once without additional printing time. This requires properly placing each copy or component within the build chamber dimensions while optimizing space and ensuring separate parts do not meld together when curing.

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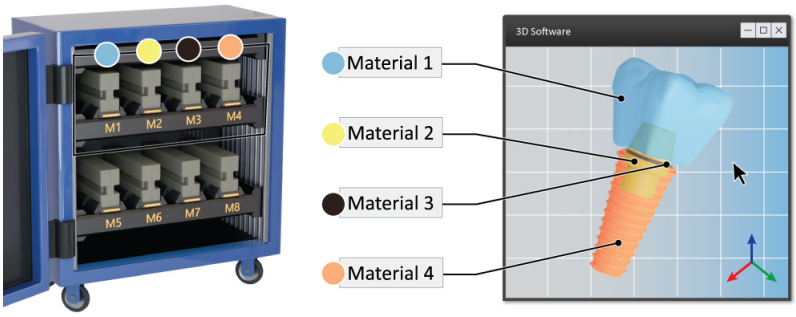
Material Selection



Setup for material jetting involves selecting and loading the correct material cartridges or tanks for build and support materials. You may also need to change which materials are selected in the machine's software using the control, or human-machine interface (HMI). The software may automatically adjust the temperature or other settings based on the selected materials. These settings must match the materials that are loaded in the material delivery system. Temperature settings ensure that build and support materials have the correct viscosity to dispense accurately.

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Material Delivery System Calibration

While material jetting machines calibrate automatically based on the print program and the selected materials, you must run verification processes to ensure all nozzles and other material delivery components are working properly for both build and support materials. This could mean printing sample layers to check whether each nozzle is dispensing materials correctly.

Some machines have a setup procedure for weighing the dispensed material in order to self-calibrate and optimize nozzle deposition quantities. Running test prints may also be used to check nozzle accuracy.

During setup, you may also need to run the machine's nozzle cleaning procedure to recalibrate the print nozzles and clear out any cured materials left from a previous build.

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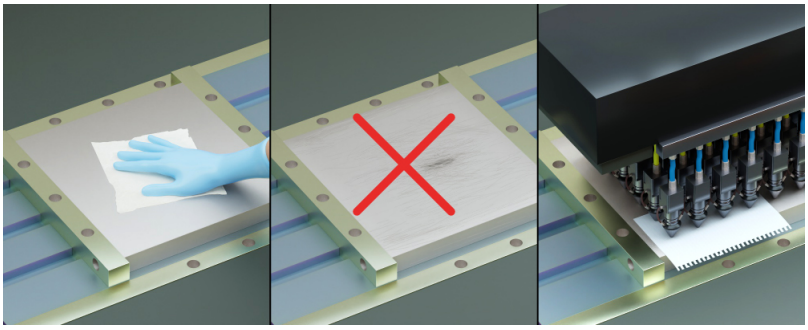
Build Platform Setup and Calibration



During setup, the platform must be clean and free of cured materials or other contaminants. The platform surface should also be inspected for damage. These conditions can prevent the materials from adhering to the surface properly. Remove any cured material from previous builds with a scraper and clean the build surface thoroughly before starting a print.

While the elevator and build platform calibrate automatically when selecting the home position on the machine, you may need to adjust the build platform to ensure it is at the appropriate level in the start position. The machine should be stationed in a level area of the facility using a leveling device. Additionally, use a strip of paper or other device to ensure the appropriate gap between the print head and the build surface.

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Review: Setup and Calibration

- **Build Orientation**: May require adjusting during setup to minimize material waste.
- **Temperature**: May require adjusting to maintain the correct viscosity.
- **Material Delivery System**: Must be run manually with the control during setup to clean and recalibrate.
- **Build Platform**: Must be set level in the start position during setup.

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Post Processing Setup

Unlike many AM processes, material jetting produces parts with high dimensional accuracy and good surface quality, requiring very little surface finishing. However, you must set up machines and equipment for removal of any support materials used or to improve the accuracy of part features requiring strict tolerances. Water jetting, chemical bath solutions, or other methods may be set up for removing wax support structures from a photopolymer resin build. Grinding and CNC machining centers can also be set up to remove build materials and add additional features if needed.

Although photopolymer resins cure with UV light during the build process, metal and ceramic materials may require additional heat treatment to properly cure or to form stronger molecular bonds.

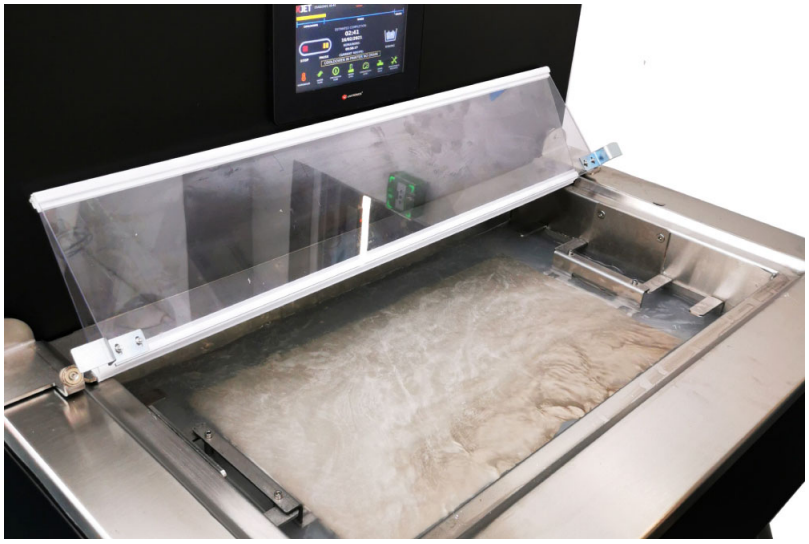

Figure 1: Support Removal (Photo courtesy of XJet)

Figure 2: Heat Treatment (Photo courtesy of XJet)

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Material Jetting Safety

Like other AM processes, material jetting presents safety risks. Operators should wear protective gloves to prevent burns or chemical exposure when handling hot resin tanks, reservoirs, or material cartridges. Proper ventilation is required to mitigate the risk of fume inhalation, especially when thermal bonding is required. Exposure to UV radiation also poses a risk of eye and skin damage. Never handle print head components or look directly into the UV lamp when the machine is in operation.

Handling metal materials requires additional personal protective equipment (PPE) to prevent metal particle inhalation. Chemical-resistant gloves and respirators can protect against respiratory exposure risks when cleaning. When working with nanoparticle materials, you should wear an appropriately rated respirator according to the National Institute for Occupational Safety and Health (NIOSH) classifications.

The manufacturing employer must supply the appropriate PPE for personnel who operate material jetting machines. First aid should be readily available, in addition to established emergency procedures, in case of accidental ingestion or inhalation. You must always read safety labels, symbols, and warnings on all equipment and have Safety Data Sheets (SDSs) readily available.

Figure 1: Nanoparticles can be transported in oils and other fluids.

Figure 2: Photo courtesy of Stratasys

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Final Review

Which machine type would be set up to remove support structures after material jetting?

- **Water jetting machine**
- Metal jetting machine
- FDM machine
- PBF machine

What is a safety risk associated with material jetting machines that use photopolymer build materials?

- **UV radiation**
- Metal particles
- Intense heat

Who is responsible for providing the correct personal protective equipment for handling hazardous materials used in manufacturing?

- **The employer**
- The material supplier
- Machine operators
- OSHA safety auditors

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Class Vocabulary

Additive Manufacturing AM. The process of joining or solidifying materials to make an object based on a three-dimensional (3D) computer model. Additive manufacturing methods typically build up layers of material to create an object.

AM Additive manufacturing. The process of joining or solidifying materials to make an object based on a three-dimensional (3D) computer model. AM methods typically build up layers of material to create an object.

Automated Run by a preprogrammed mechanical system with little to no human intervention. Automated machines operate more efficiently and precisely than machines that an operator manually controls.

Axis An invisible straight line that helps describe the movement of an object in three-dimensional (3D) space. On material jetting machines, each axis is aligned with the machine coordinate system.

Build Chamber The area contained within an additive manufacturing (AM) machine where the part is constructed. Many build chambers are enclosed to improve the safety of the manufacturing operation.

Build Orientation The position in which a part is placed on a build platform during an additive manufacturing (AM) process. Build orientation, or part orientation, involves balancing part quality and build times.

Build Platform The flat surface on which a part is additively manufactured. The build platform can either be a permanent machine surface from which parts are removed or a surface that can be removed from a machine once the build is complete.

Calibrated Compared and adjusted to match a known, accurate standard. Machines are calibrated to eliminate any unwanted variation.

Ceramic A hard, brittle material that can withstand high temperatures and resist corrosion. Ceramics, which include metal oxides, nitrides, and glasses, are sometimes used as build material in additive manufacturing (AM).

Chemical Bath A specialized solution that an engineer dips an additive manufacturing part into in order to remove support structures. A chemical bath is formulated to dissolve support structures without harming the part.

CNC Machining Centers A sophisticated computer numerical control (CNC) machine that can perform multiple machining operations in the same setup with a variety of tools. CNC machining centers can be effective for post processing of parts produced with metal additive manufacturing (AM) methods.

Computer-Aided Design CAD. Computer software used to create a 3D model of a part before it goes into prototyping or production. Computer-aided design models are digital representations of the different aspects of a part.

Contaminants Any foreign substance, such as dirt, that can reduce efficiency of machines and damage components. Contaminants can cause machines to wear prematurely and reduce service life.

Cure To cause a material to bond and solidify by permanently cross-linking its molecules. Curing techniques for photopolymers include exposing the material to ultraviolet light during the production process.

Elevator The device on a material jetting machine that raises and lowers the build platform along the Z axis. The elevator may need to be jogged or calibrated to ensure proper leveling of the build surface.

End-Use Designed to be used directly by a consumer or directly in another manufactured product. End-use products created by additive manufacturing (AM) include medical implants, custom dental devices, and camera equipment.

Features A distinguishing characteristic that performs a function on a part. Features include grooves, shoulders, and hinges, among many others.

Gantry The mechanized bridge upon which tooling is mounted. The gantry on a material jetting machine moves along the X axis or along the X and Y axes, depending on the machine.

Geometry The linear and curved shapes that characterize a part. Complex part geometry on material jetting machines requires the use of support structures.

Grinding A machining process in which an abrasive is used to precisely cut and shape the surface of a workpiece. Grinding operations are often used to remove small amounts of material from a workpiece and create good surface finish.

Heating Lamp A device that is used in specialized material jetting machines that produces heat in order to cure materials. Heating lamps are equipped on nanoparticle jetting (NPJ) machines.

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Home Position The default position on an additive manufacturing (AM) machine that is typically the farthest possible distance from the start position. The home position on material jetting machines is the default position of the print head in the X and Y axes and the elevator in the Z axis.

Human-Machine Interface HMI. A device connected to a machine that allows operators to monitor and control a machine or process. A human-machine interface may have a touch screen or control pad.

Leveling To keep the surface of one or more components completely horizontal and in alignment. Leveling the build platform in material jetting ensures that a part is constructed accurately.

Liquid Metal Jetting LMJ. An advanced material jetting process that builds parts using molten, liquid metal. Liquid metal jetting, unlike nanoparticle jetting (NPJ), creates parts without a binding agent.

Material Delivery System The collection of components on an additive manufacturing machine that transport the build materials onto the build surface. The material delivery system on a material jetting machine includes the material cartridges, pneumatic or piezoelectric actuation mechanism, delivery tubes, and print head nozzles.

Material Jetting An additive manufacturing process in which droplets of build material are selectively deposited onto a build platform and cured by ultraviolet (UV) light or other methods. Material jetting generally includes Polyjetting, which uses photopolymer resins, but can also include nanoparticle jetting and liquid metal jetting methods.

Molecular Bonds The joining of atoms within molecules of a material. Molecular bonds can be influenced by temperature and other conditions.

Nanoparticle Jetting NPJ. A material jetting process that builds parts using ceramic or metal nanoparticles suspended in a binder that evaporates. Nanoparticle jetting can potentially create parts designed for heavier use than photopolymer material jetting.

Nozzles A spout at the end of piping or tubing through which substances are funneled. Nozzles are used to distribute build material during some additive manufacturing processes.

Overhangs A section or feature on a part that extends outward and past the base. Overhangs are often created using support material in material jetting.

Personal Protective Equipment PPE. An article of clothing or a device worn to minimize exposure to hazards and prevent injury. Personal protective equipment may include safety glasses, safety gloves, ear plugs, respirators, or steel-toed boots.

Photopolymer A thermoset polymer that cures and hardens when exposed to ultraviolet (UV) light. Photopolymers are used in material jetting.

Photoreactive Resin A material that chemically reacts to ultraviolet (UV) light when exposed. Photoreactive resins solidify and harden when exposed to UV light.

Piezoelectric Having the ability to generate an electrical charge when subjected to mechanical pressure or stress. Piezoelectric actuation is used in drop-on-demand inkjet technology.

Pneumatic A type of power created by the controlled motion of a gas, usually air, under pressure. Pneumatic systems are often used to power robotic devices and are also used in additive manufacturing machine components.

Post-Processing A procedure used to clean, improve, or otherwise finish a part after it has been manufactured. Post-processing procedures include support removal, abrasive finishing, heat treatment, and painting.

Print Head The moving component on a printing machine that holds and distributes the part build material. Additive manufacturing (AM) machines, such as those used in material jetting or binder jetting, have print heads.

Print Resolution The fineness of detail in a computer-aided design (CAD) model that determines the accuracy of a printed part. Print resolution helps determine the part's surface finish quality, with greater resolution leading to better surface finish.

Print Speed The rate at which a print head or worktable moves in a linear or rotational axis while printing. Print speed is typically measured in millimeters per second (mm/s).

Prototyping To create an early sample, model, or release of a product built to test a concept or process. Photopolymer material jetting is primarily used for prototyping.

Quenching The controlled cooling of a metal after it has been heated to very high temperatures during heat treatment processes. Quenching rates are determined by the metal type.

Recalibrate To readjust or reset a machine's calibration after initial setup. Recalibrating helps eliminate variation between operations and during calibration testing.

Reservoir A container used to store large quantities of liquid materials. Reservoirs are used in liquid metal jetting to store molten metal.

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Resin Tanks An industrial receptacle used to hold fluids. Resin tanks are used to hold liquid photopolymers in material jetting.

Respirators A breathing device worn to prevent inhalation of hazardous substances. Oil-resistant respirators should be used when cleaning nanoparticle materials since nanoparticles can be transferred through fluids, including the bloodstream.

Roller A cylindrical device used in additive manufacturing to either deposit and level build material or fuse layers of a part together. Rollers are used to level material in material jetting, vat photopolymerization, binder jetting, powder bed fusion (PBF), and sheet lamination methods.

Safety Data Sheets SDS. The mandatory information that must accompany almost every chemical in the workplace. A Safety Data Sheet includes details such as the hazards, precautions, and first aid procedures associated with the chemical.

Software The programs, instructions, and formulas that structure the actions of a computer. Software controls computer hardware functions and operations.

Start Position The location where the operator places the build platform during material jetting. The start position is considered the best position for a print.

Support Structures A reinforcing component used to hold the weight of an additively manufactured part as it is being constructed. Support structures are removed from the part once the build is complete.

Test Prints A procedure performed during the additive manufacturing process to assess the performance of the machine. Test prints are often used to calibrate the accuracy of the print head nozzles, UV light, or machine calibration.

Thermal Bonding The use of different temperature conditions to join molecules together within materials. Thermal bonding can join different types of materials together, such as plastic to metal or ceramic.

Three-Axis A type of machine control that can execute movement along three directions concurrently. Three-axis machines move machine tools along three linear axes.

Three-Dimensional 3D. Having height, width, and depth. Three-dimensional parts are created during additive manufacturing (AM) processes.

Tolerances The unwanted but acceptable deviation from the desired dimension. Products must be within specified tolerances to function properly.

Toolpaths The series of coordinate positions that determine the movement of a tool during a manufacturing operation. Toolpaths for additive manufacturing machines are automatically generated by specialized slicing computer programs that analyze the 3D part model and divide it into layers.

Troubleshooting A systematic approach to solving problems quickly and efficiently. Troubleshooting often involves a logical process of elimination to identify the source of a problem.

Ultraviolet Light UV light. A potentially harmful wavelength of light that is below or shorter than violet on the light spectrum. Ultraviolet light is used to selectively solidify photopolymer in material jetting.

UV Ultraviolet. A potentially harmful wavelength of light that is below, or shorter than, violet on the light spectrum. UV light is used to selectively solidify a photopolymer in material jetting.

Ventilation A means of circulating clean, fresh air. Ventilation ensures that air containing potentially harmful dust or gas is removed from a space and replaced with clean air.

Viscosity A fluid's resistance to flow. Temperature settings on material jetting machines affect the viscosity of the resin materials.

Water Jetting A material removal method that uses a powerful, high-velocity stream of water to cut or remove materials. An abrasive can be added to the water to facilitate the water jetting process.

Wire Feed System A machine or component that delivers material feedstock in the form of a wire spool. Wire feed systems may need to be calibrated or changed during setup.

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